

EE 3201.302

Lab 8 Diodes

Krish Dhage: EE

Kaushal Sundaramoorthy: EE

Dr. Tooraj Nikoubin

TA: Dipali Deepak

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I. Abstract

This lab focused on constructing various diode-based circuits—including a limiting circuit, half-wave rectifier, full-wave rectifier, and power supply—and comparing their real-world behavior to simulations performed in PSpice. Proper orientation of diodes and accurate wiring were critical to replicating the expected waveforms. Using the Analog Discovery 2 (AD2) and Waveforms software, each circuit was built and tested, with results closely matching the corresponding PSpice simulations. The successful replication of the simulated waveforms in the physical circuits demonstrates a strong correlation between theoretical predictions and practical implementation.

II. Introduction

A diode is a semiconductor device that permits current flow in one direction, effectively acting as a “one-way switch” for electrical signals. This property enables key functionalities such as rectification (AC to DC conversion), signal protection, and voltage regulation. Zener diodes, a special type of diode, are designed to operate in reverse bias within the breakdown region, making them useful as voltage regulators or reference components. Diodes are widely used in power conversion applications, allowing AC power to be converted into DC.

In this lab, understanding and interpreting the forward and reverse characteristics of diodes was essential. Approximations were used in the prelab to estimate the voltage drop across diodes, and bias analysis was applied to solve circuits iteratively. Simulations in PSpice were used to graph the I-V characteristics of diodes, which were then compared to actual measurements of voltage and current taken during the lab to evaluate the accuracy of the theoretical models.

Four circuits were built and analyzed: a limiting circuit, a half-wave rectifier, a full-wave rectifier, and a voltage regulator. These circuits were constructed and simulated using waveforms to observe their output waveforms. The I-V characteristic of a component represents the relationship between current (y-axis) and voltage (x-axis), while the transfer function of a circuit plots output voltage (V_{out}) against input voltage (V_{in}). Overall, this lab demonstrated the practical applications of diodes and how various configurations can achieve different circuit behaviors.

III. Procedure

1. Limiting Rectifier Circuit
 - a. Build the circuit

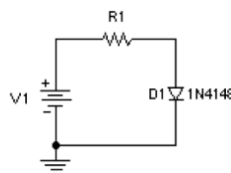


Figure 1.1: Limiting Circuit

b. Calculations of circuit values

Method 1: Hand Calculations:

Let $V_1 = 5\text{ V}$ and assume the diode's turn-on voltage to be $V_f = 700\text{ mV}$. Determine the resistor value required to set the diode current to 4.3 mA and show the required work.

We can use KVL:

$$-5\text{ V} + V_{R1} + 700\text{ mV} = 0$$

$$V_{R1} = 5\text{ V} - 700\text{ mV}$$

$$V_{R1} = 4.3\text{ V}$$

Now, because the current of components in series is the same ($I_{D1} = I_{R1}$), we can use Ohm's Law to find the value of the resistor:

$$R = V/I = 4.3\text{ V}/4.3\text{ mA} = 1\text{ k}\Omega = 1000\text{ }\Omega$$

The resistor value required to set the diode current to 4.3 mA is 1000 ohms .

Method 2 PSpice Calculations:

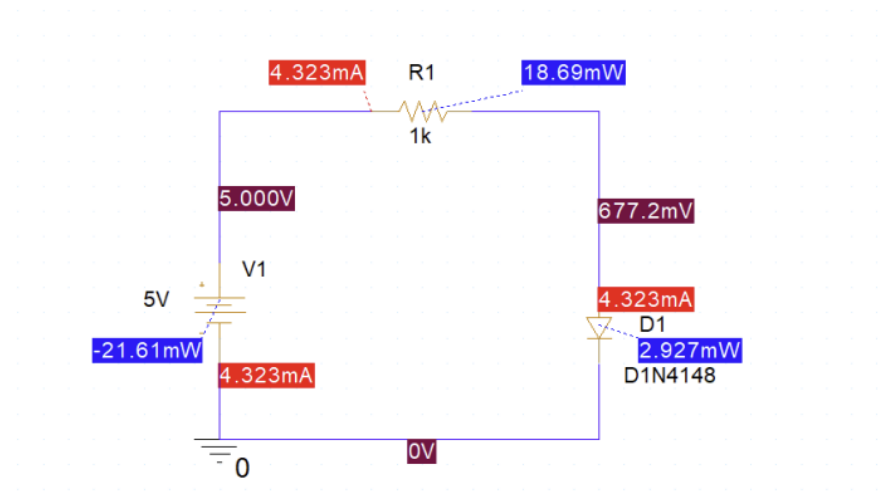


Figure 1.2: Limiting Circuit Simulation

With the calculated value of the resistor, it led to a very close estimation of the diode's required current (4.323 mA instead of 4.3 mA) and turn-on voltage (677.2 mV instead of 700 mV).

- c. Simulate I-V characteristic from prelab with the resistor load line plot on it and the diode I-V characteristic

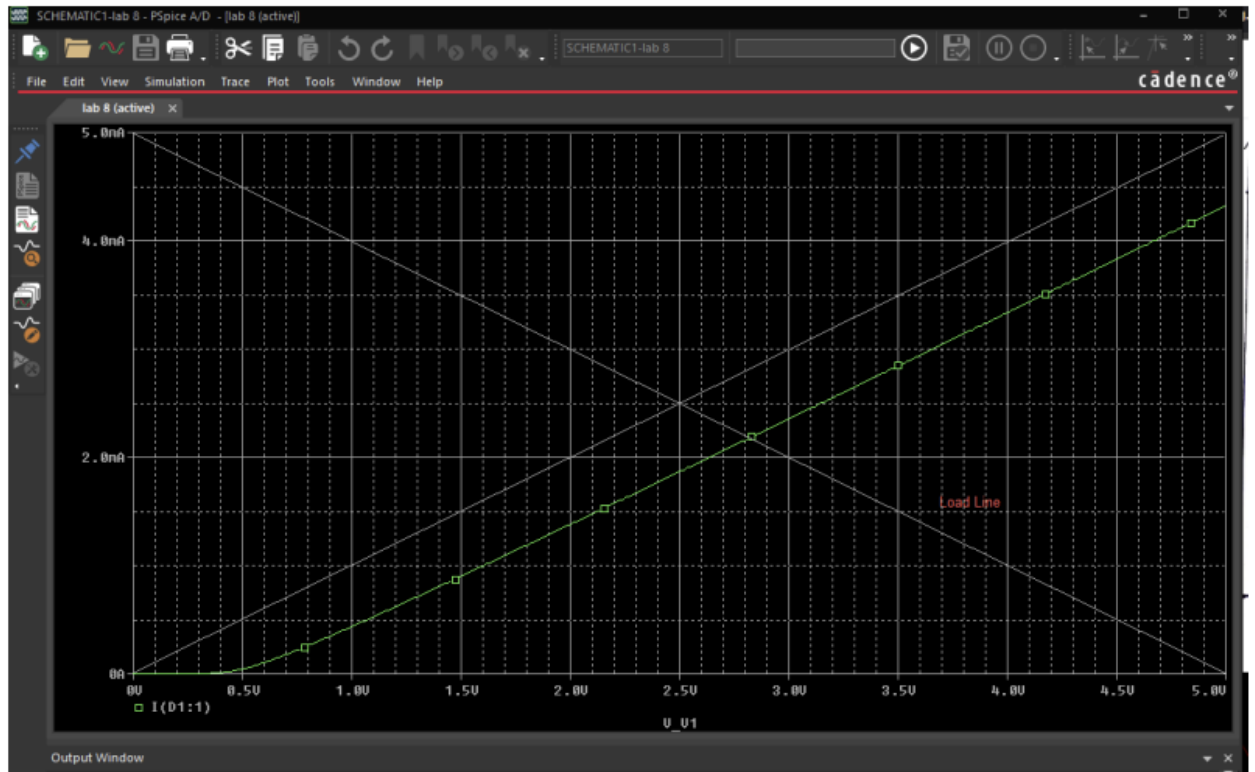


Figure 1.3: Limiting Circuit I-V Relationship

- d. Simulated Transfer Function

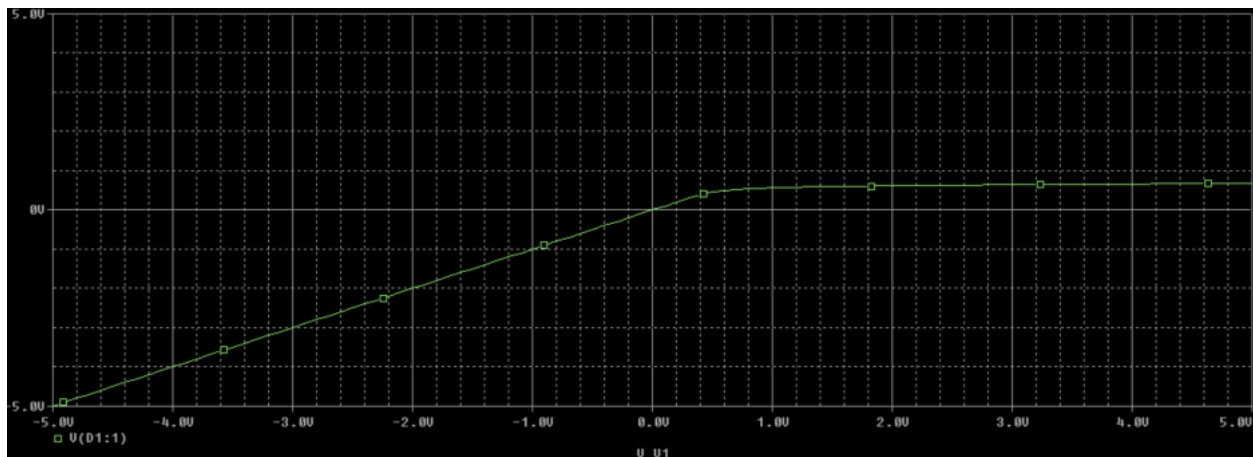


Figure 1.4: Simulation of Transfer Function of Limiting Circuit

e. Simulation of Output Waveform

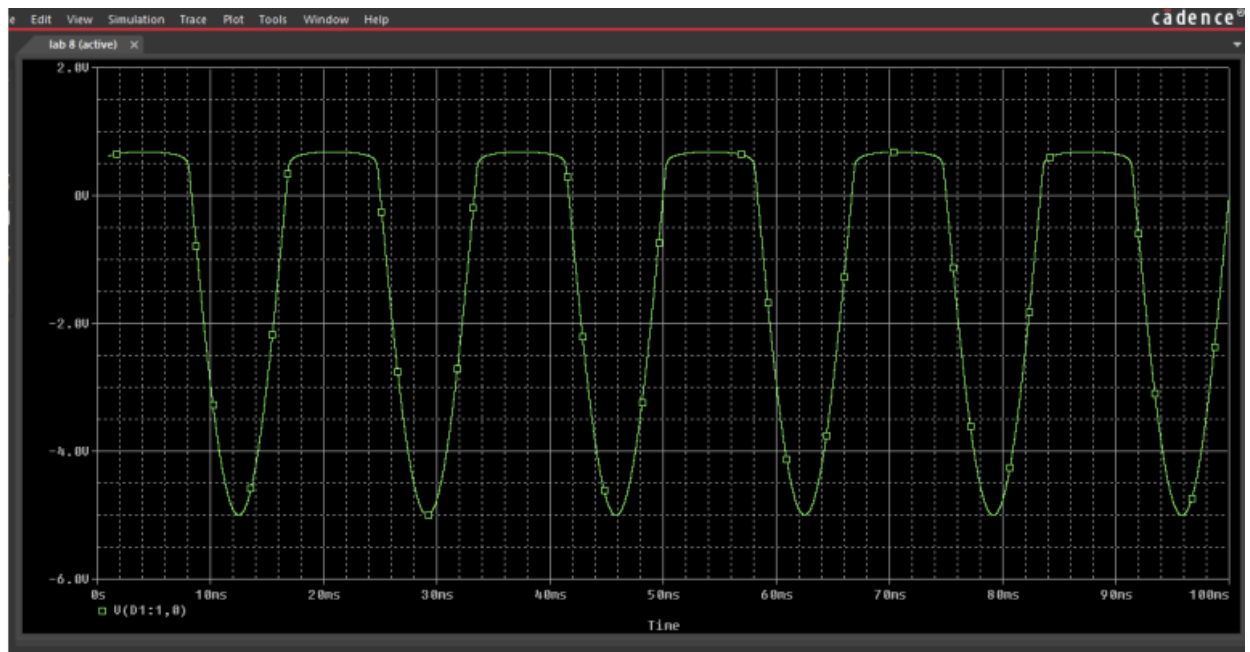


Figure 1.5: Output Waveform Simulation of Limiting Circuit

f. Physical Circuit Simulation Results

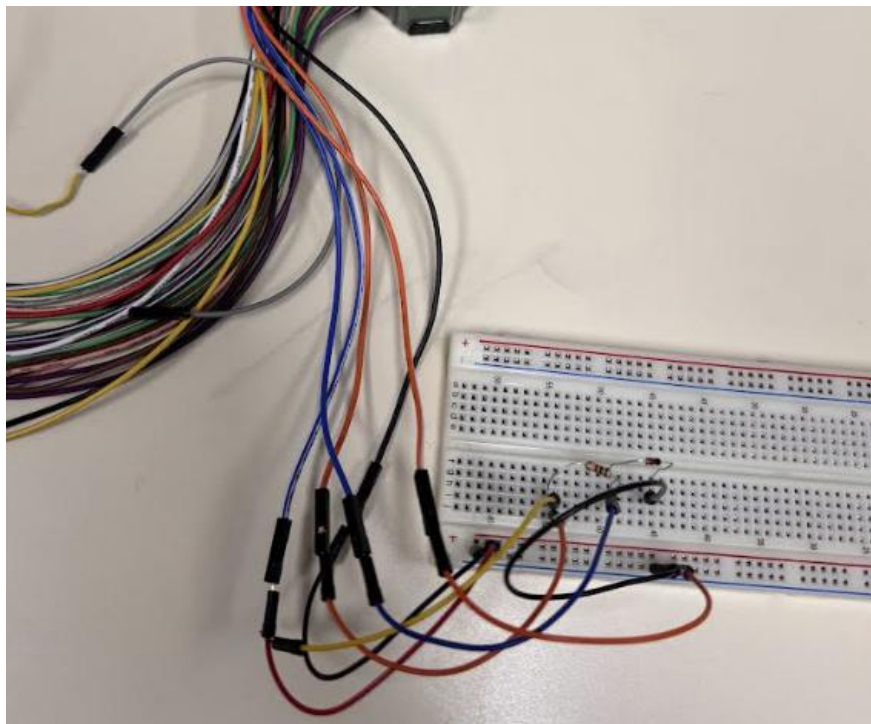
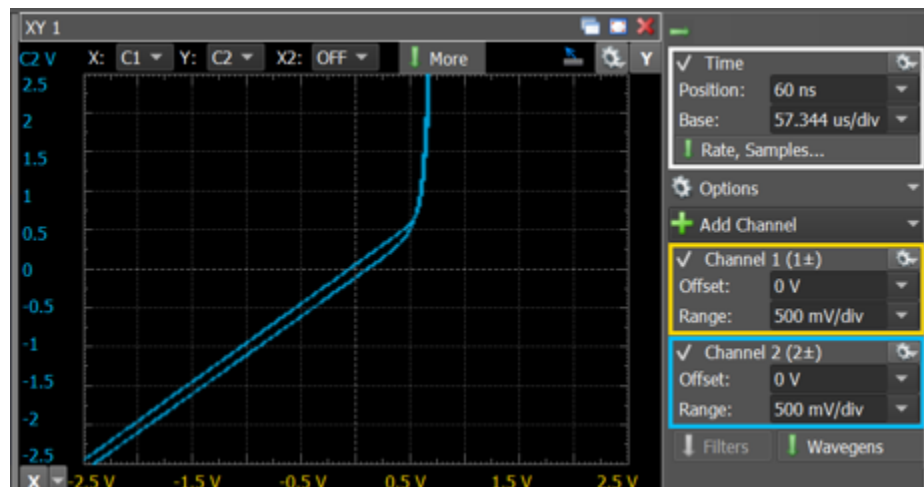
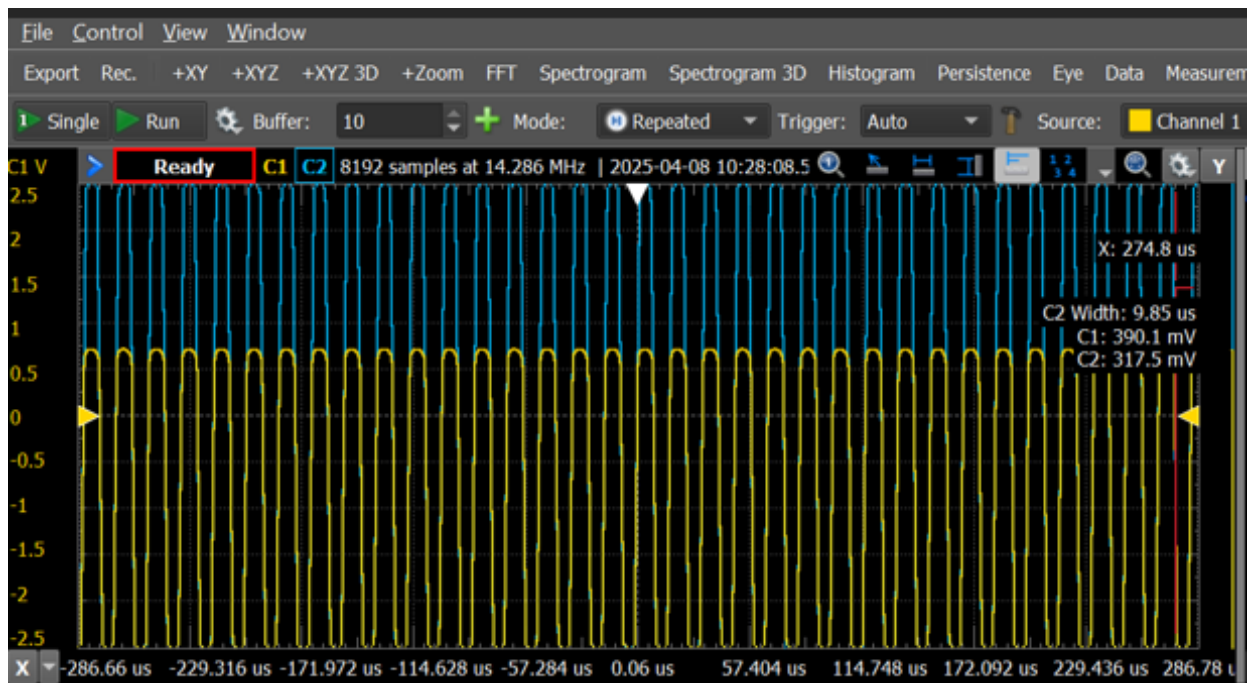


Figure 1.6: Limiting Circuit Physical Setup



g. Comparing Results

Theoretical calculations and PSpice simulations predicted the I-V characteristics of the diode-resistor pair. In the lab, a limiting circuit was built and tested with a 10 Vpp sinusoidal input. Measured output waveforms confirmed the diode effectively limited voltage as expected. The DC sweep graph showed the diode's influence on circuit behavior. Minor differences from simulations were likely due to component tolerances or oscilloscope calibration, but overall results aligned well with predictions.

2. Half-Wave Rectifier
 - a. Build the Circuit

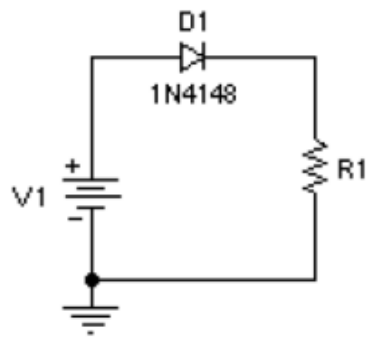


Figure 2.1: Half-Wave Rectifier Circuit Setup

- b. Transfer Function

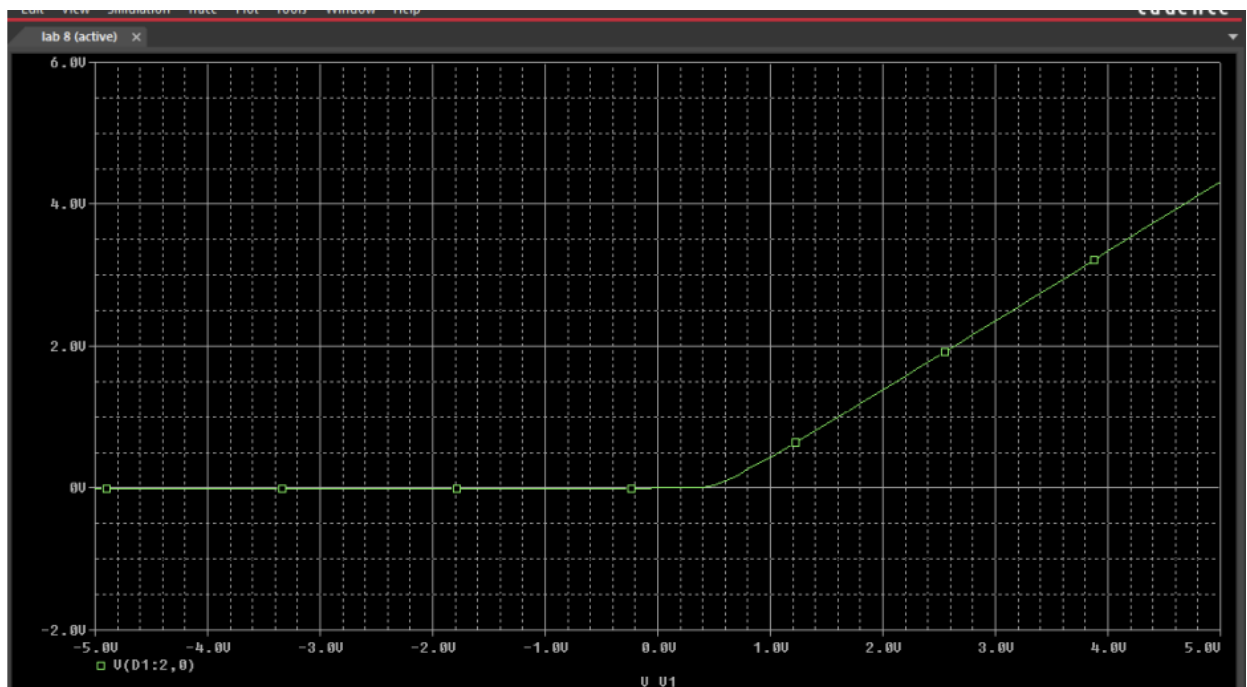


Figure 2.2: Half-Wave Rectifier Transfer Function

c. Pspice Circuit Waveform Results

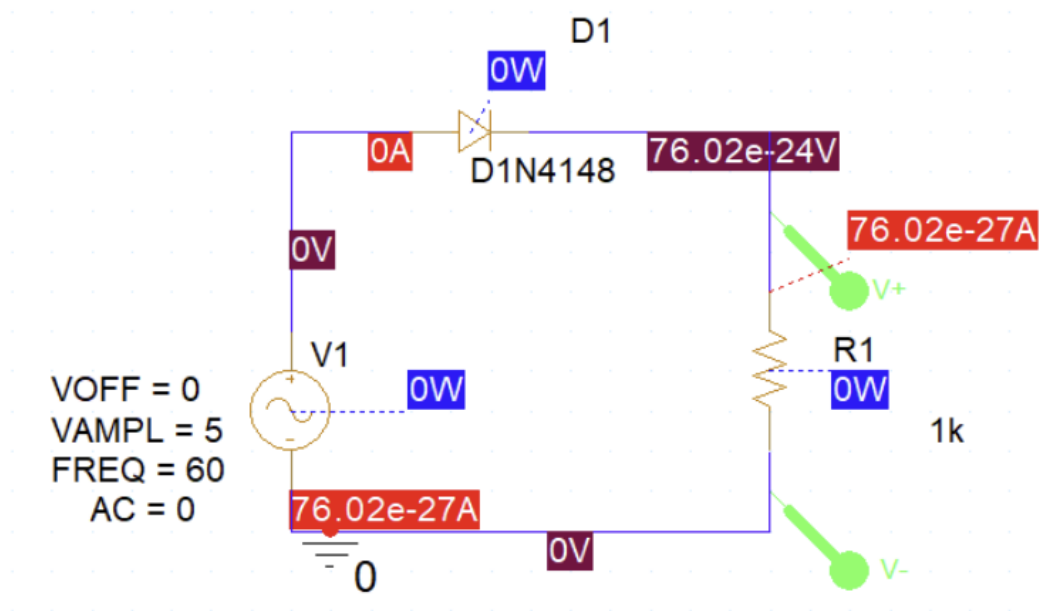


Figure 2.3: Half-Wave Rectifier Pspice Circuit Simulation

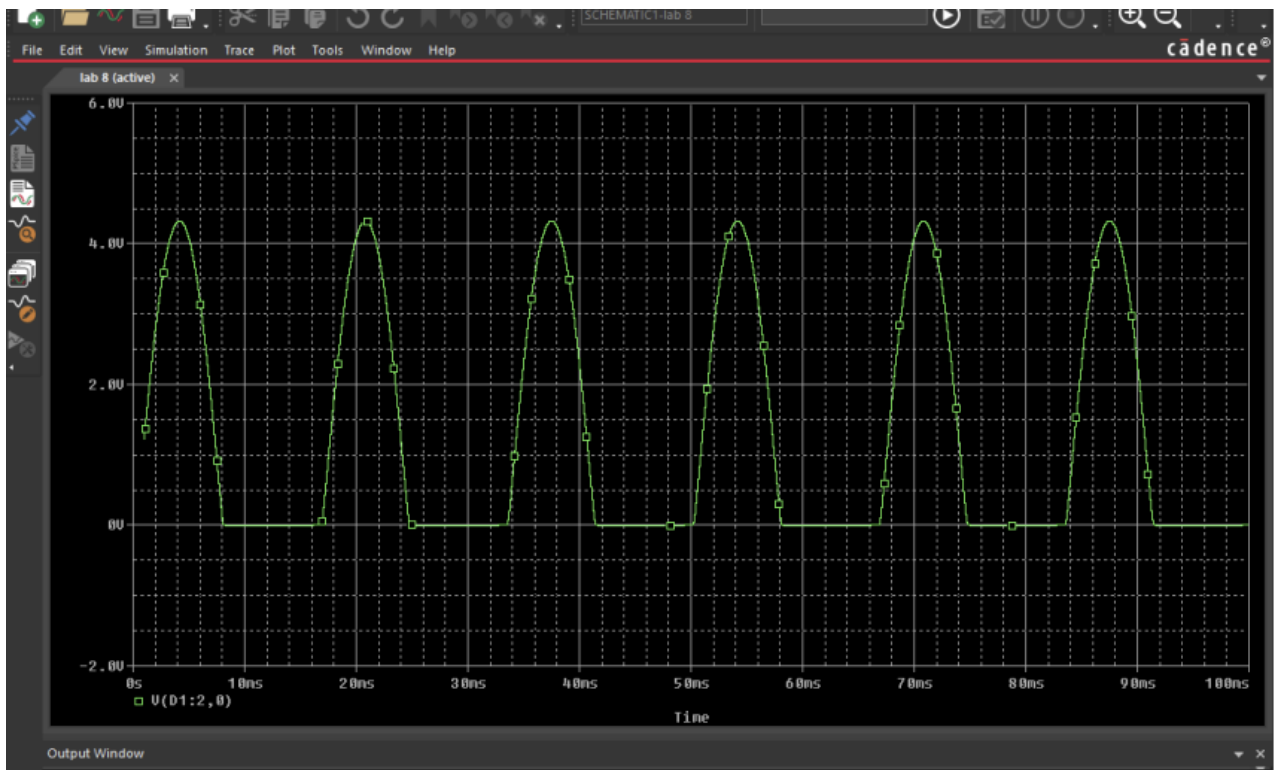


Figure 2.4: Half-Wave Rectifier Voltage Results

d. Half-Wave Rectifier Physical Circuit Results

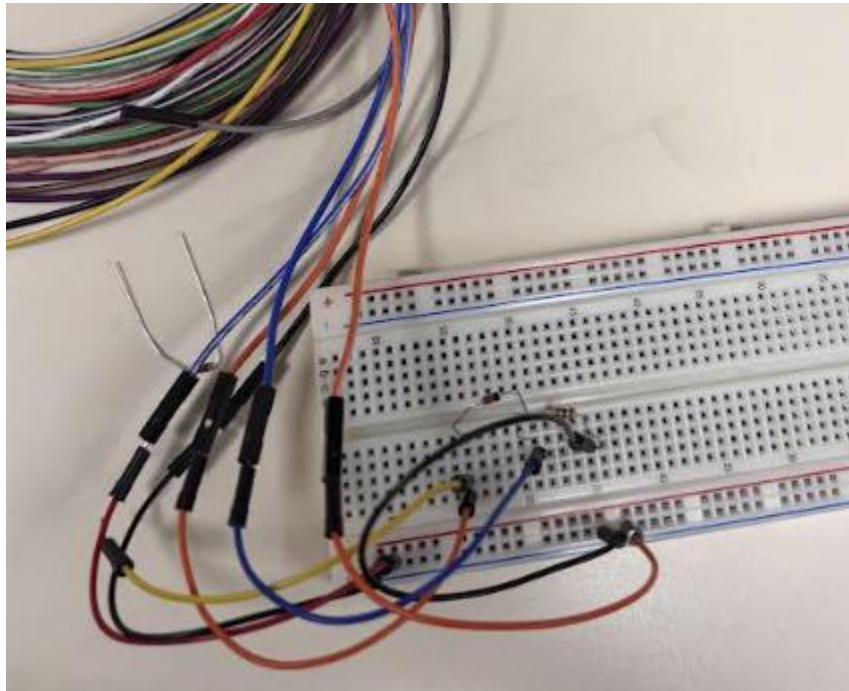


Figure 2.5: Half-Wave Rectifier Physical Circuit Setup

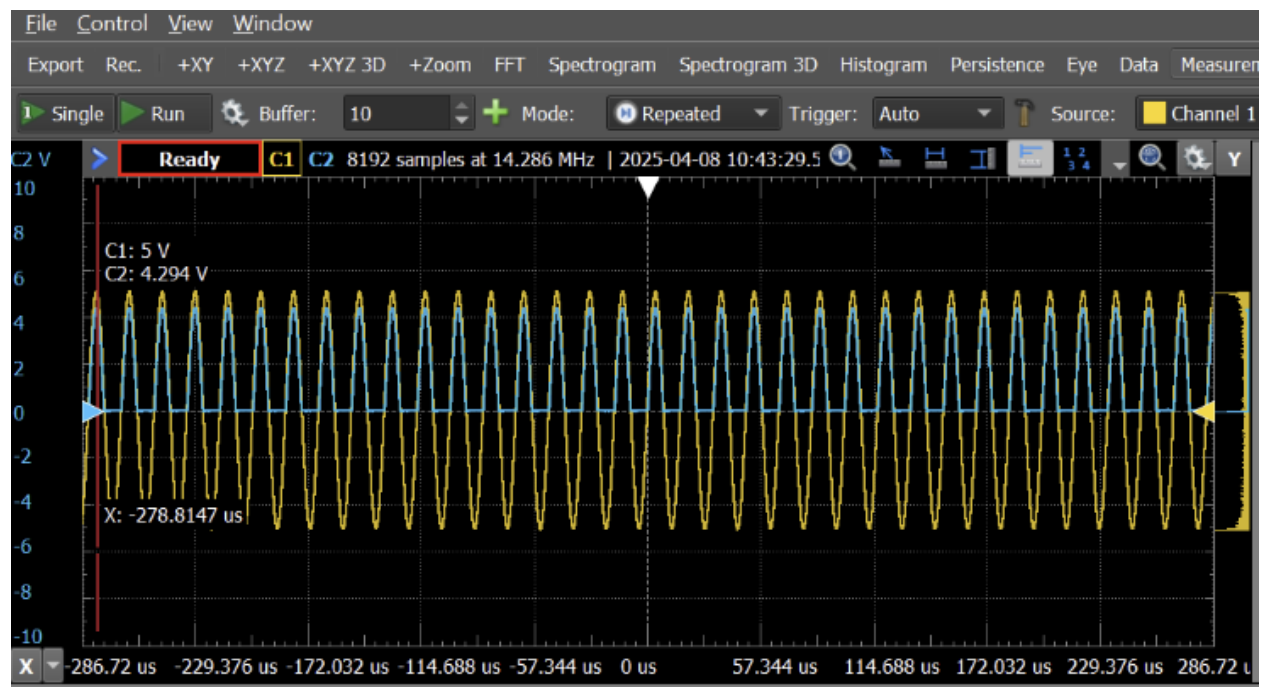


Figure 2.6: Input and Output Voltage Results

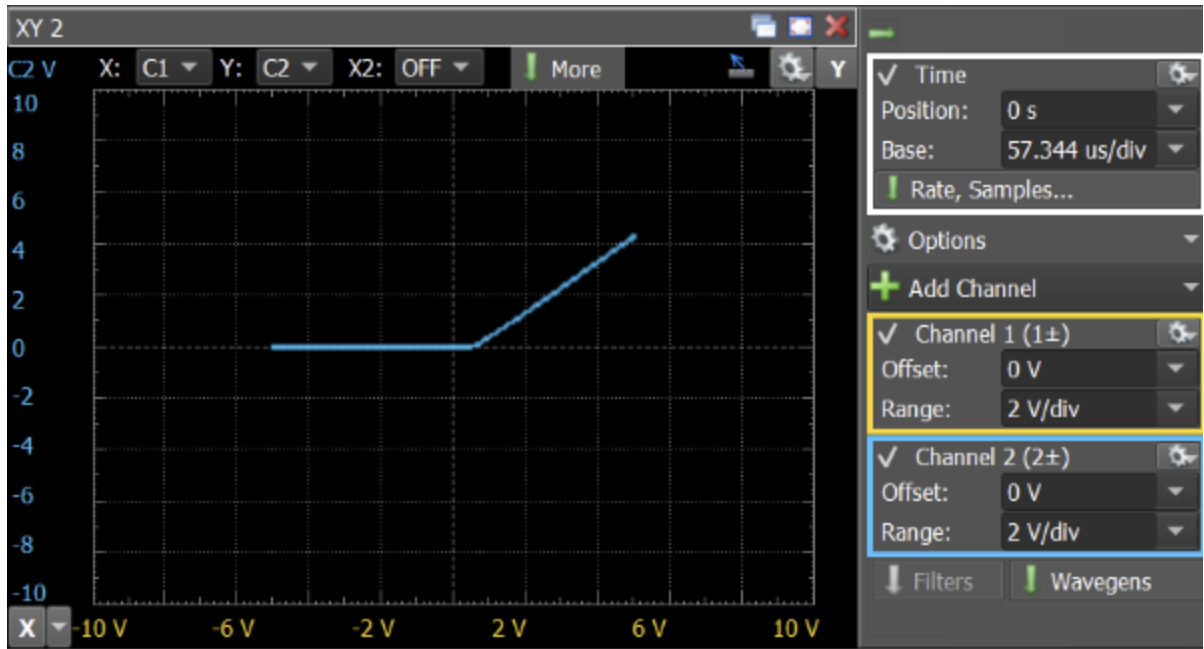


Figure 2.7: X-Y Plot of Half-wave Rectifier

e. Comparing Results

The half-wave rectifier was analyzed theoretically, predicting that only one half of the AC signal would pass, producing a DC-biased output. PSpice simulations showed a clipped sinusoid with a peak near the diode's turn-on voltage. In the lab, the circuit was built with a 1 k Ω load and a 10 V_{pp} input. Oscilloscope readings confirmed the diode blocked one half of the AC cycle, resulting in a pulsating DC output. Experimental results closely matched simulations, with minor differences due to non-ideal diode behavior. The reduced output peak further validated the diode's rectifying function.

3. Full-Wave Rectifier

a. Build the Circuit

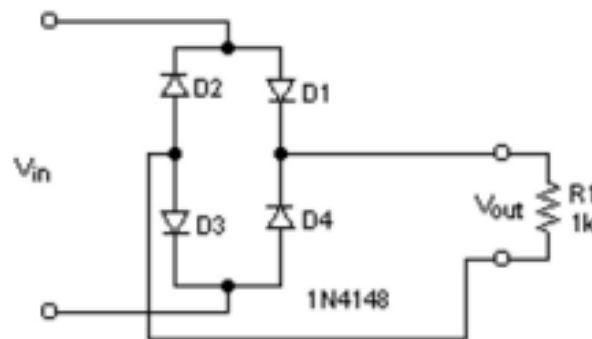


Figure 3.1: Full-Wave Rectifier Setup

b. Transfer Function

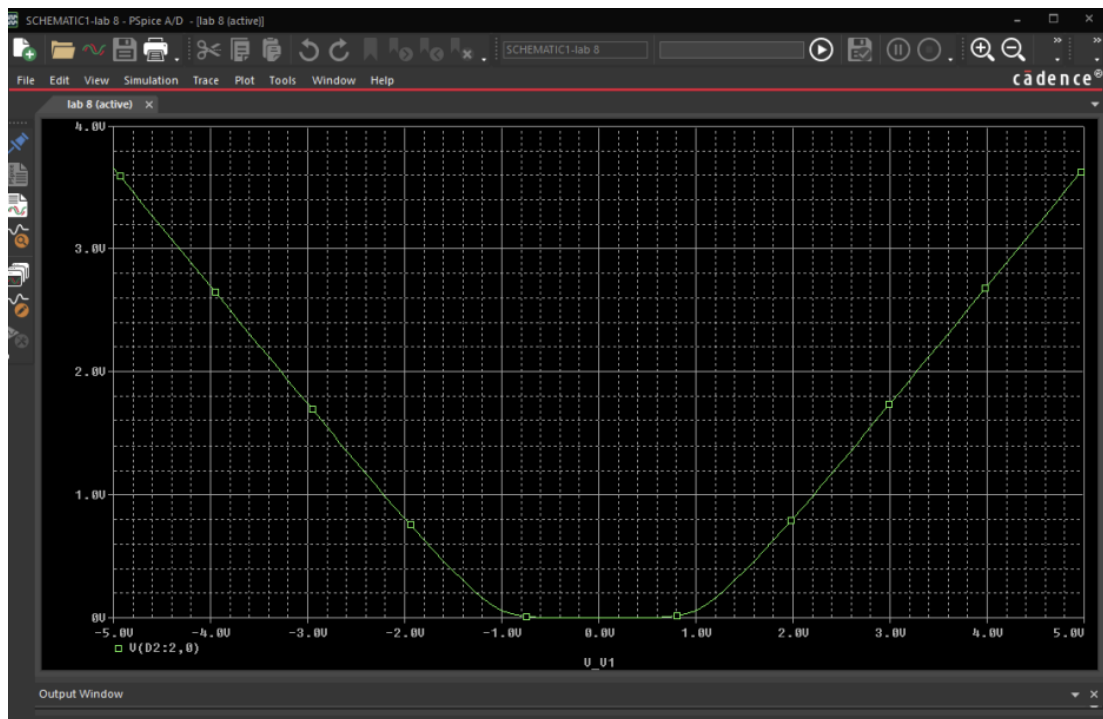


Figure 3.2: Full-Wave Rectifier Transfer Function

c. Pspice Circuit Waveform Results

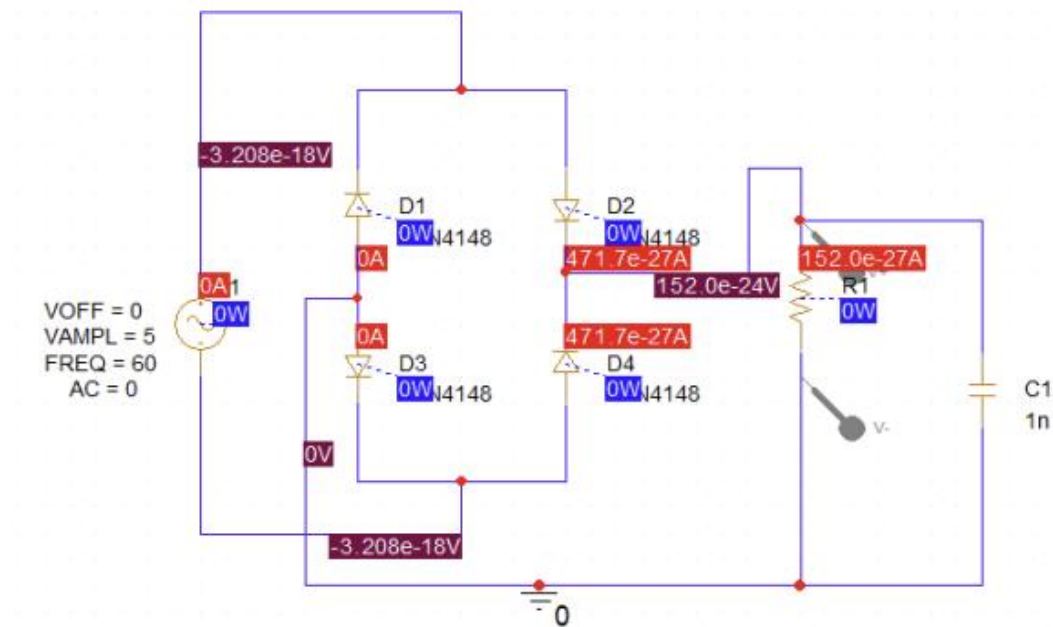


Figure 3.3: Full-Wave Rectifier Circuit Simulation



Figure 3.4: Full-Wave Rectifier Output Voltage Simulation

d. Full-Wave Rectifier Physical Circuit Results

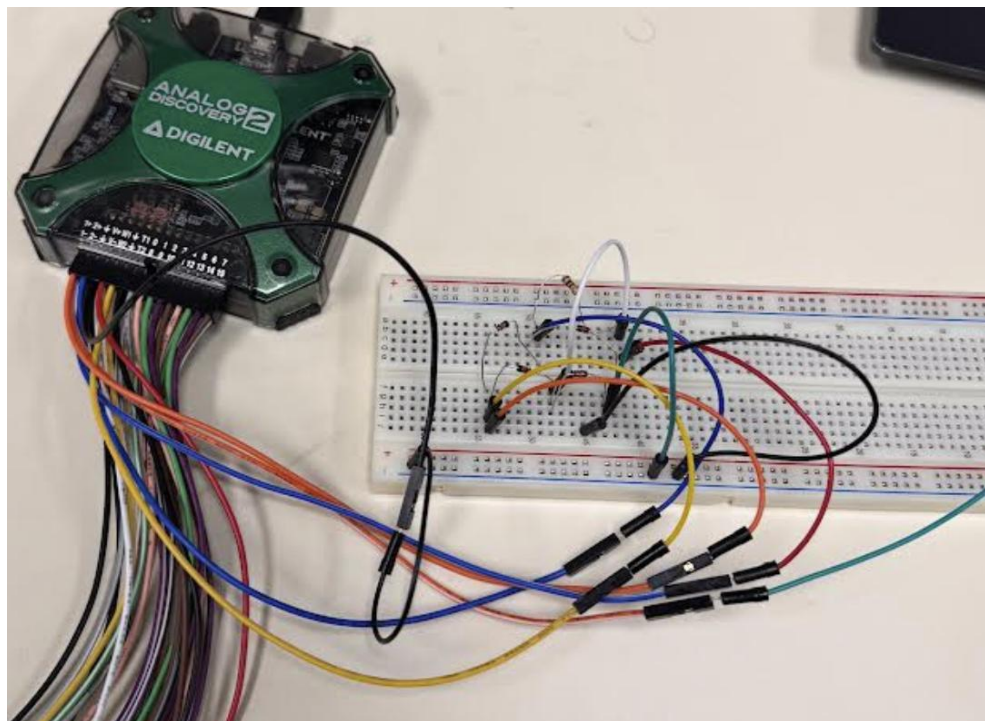


Figure 3.5: Full-Wave Rectifier Physical Circuit Setup

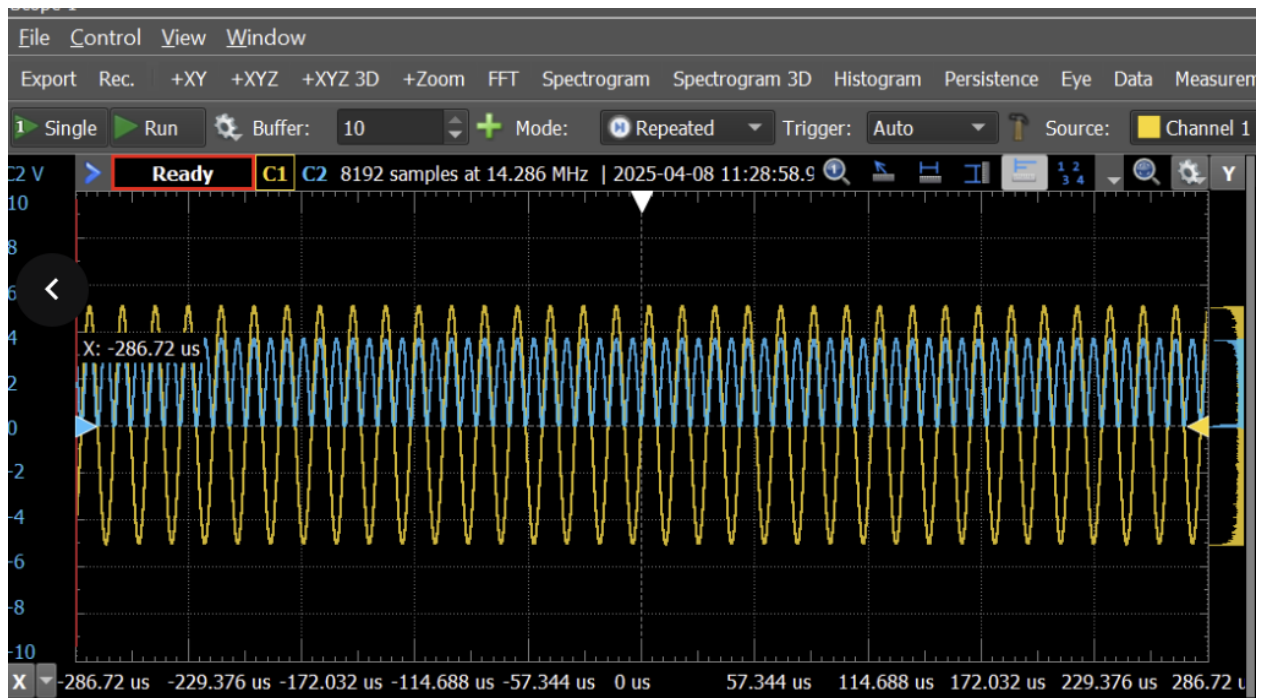


Figure 3.6: Input and Output Voltages Waveform

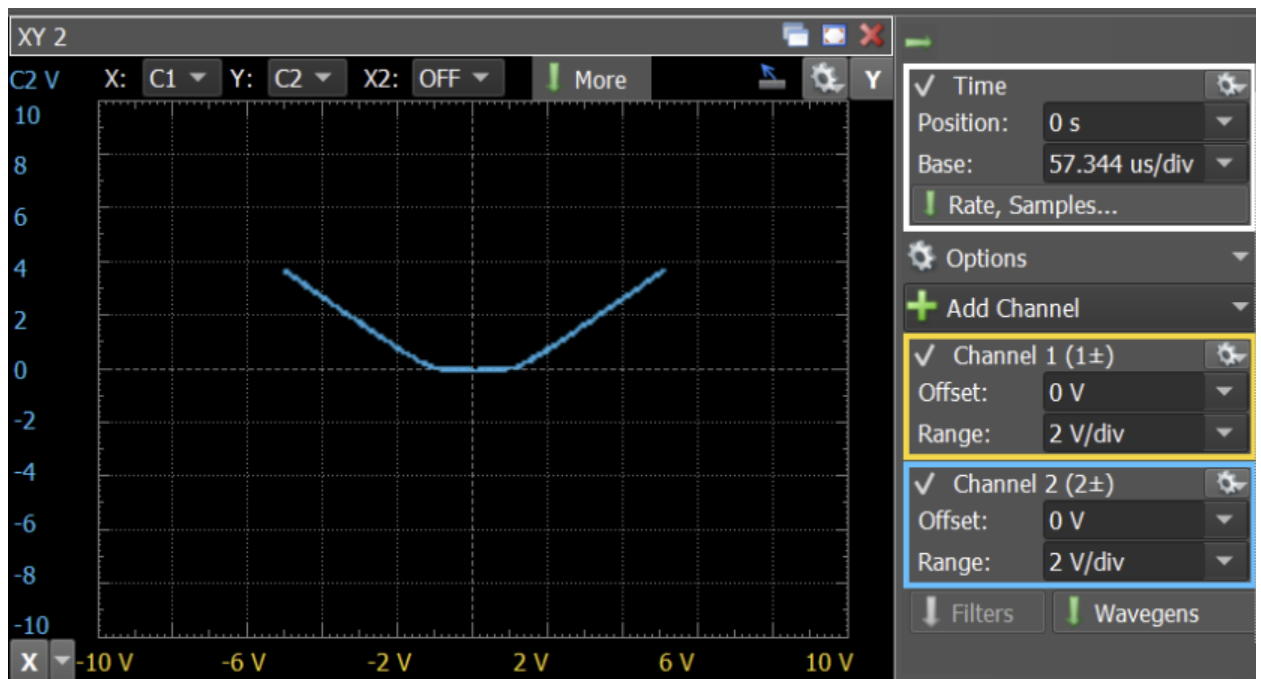


Figure 3.7: Full-Wave Rectifier Transfer Function

The simulated transfer function closely matched the one measured in the lab. The input and output waveforms observed during the experiment also aligned well with the PSpice simulations.

e. Adding a Capacitor in Parallel

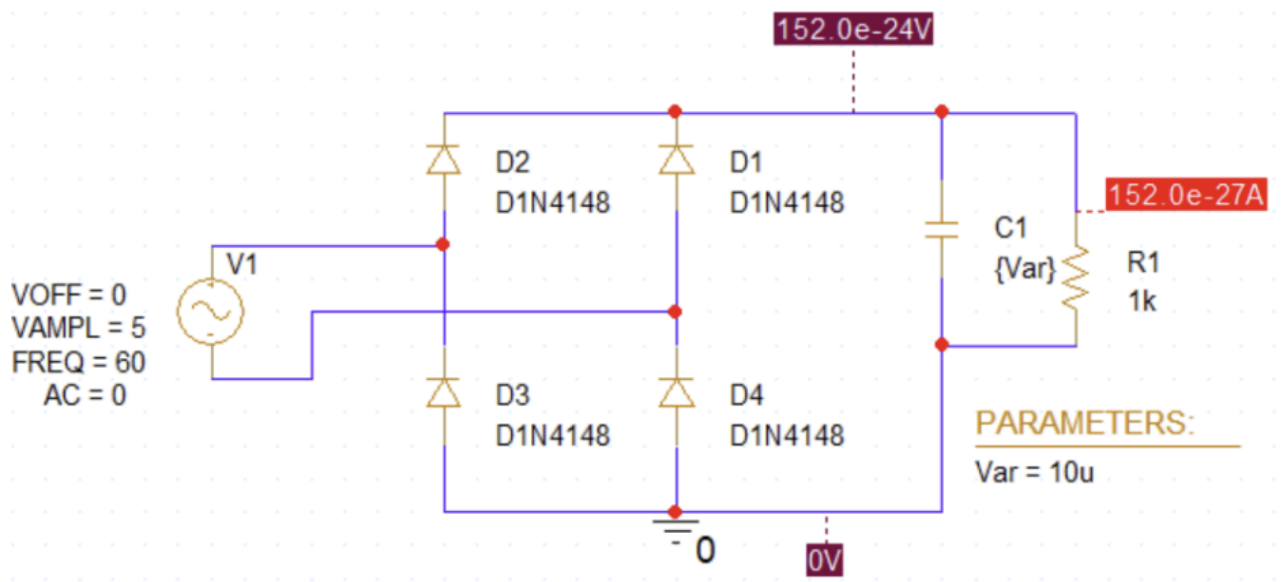


Figure 3.8: Full-Wave Rectifier with a capacitor in Parallel Schematic

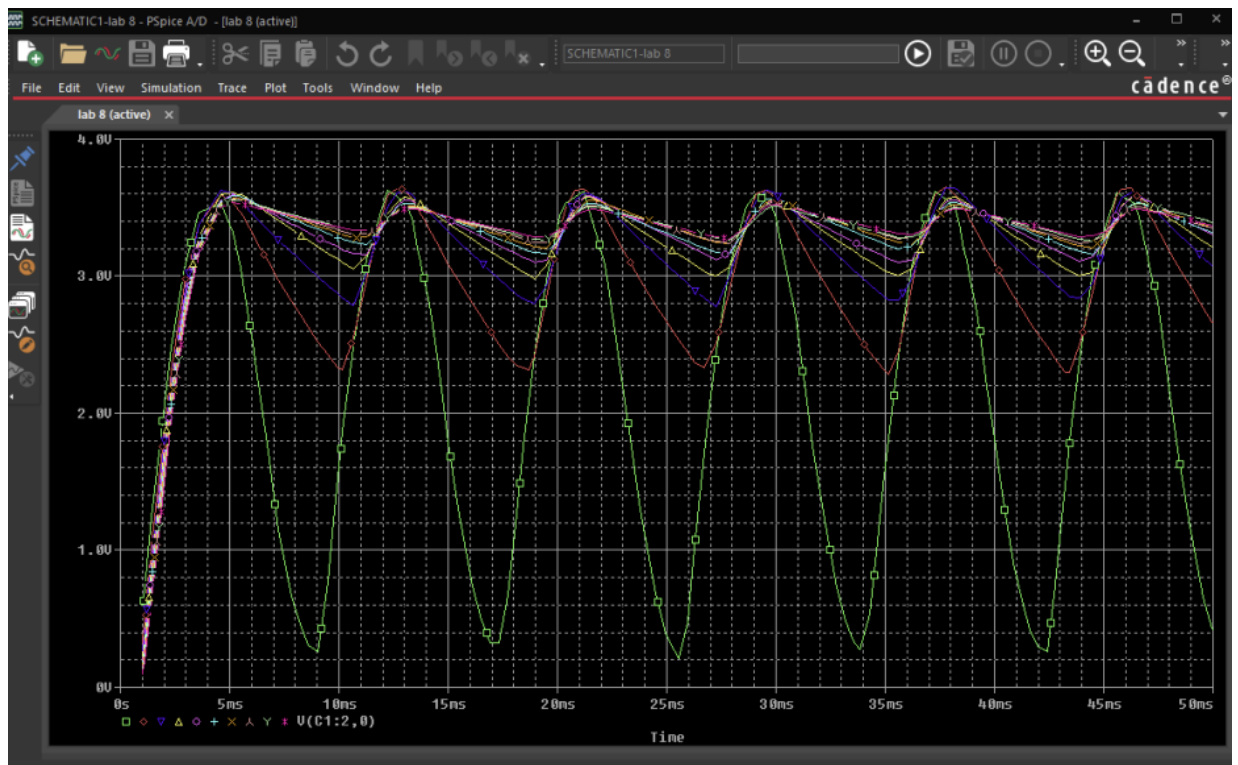


Figure 3.9: Ripple Voltage vs. Capacitance Simulation with Parametric Sweep

f. Ripple Voltage Analysis

- 1uF Capacitor

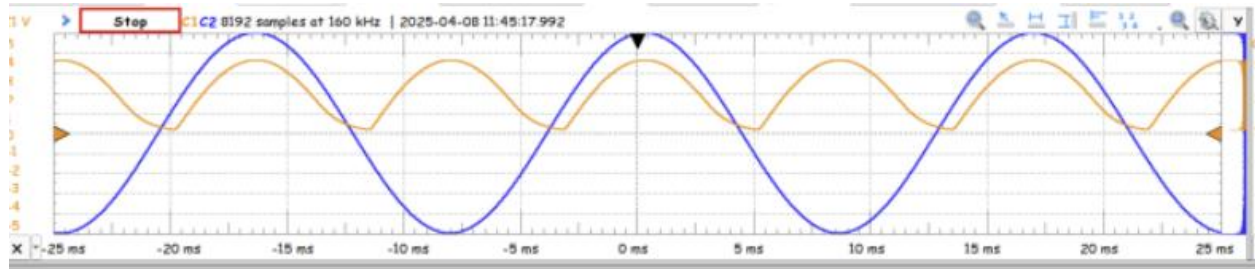


Figure 3.10: Waveform with 1uF Capacitor

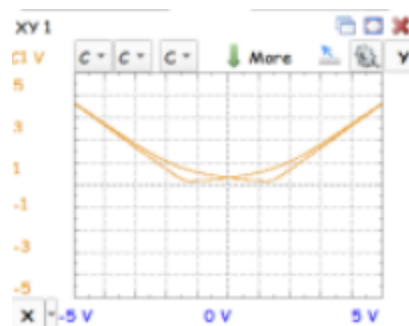


Figure 3.11: Transfer Function with a 1uF Capacitor

- 10uF Capacitor

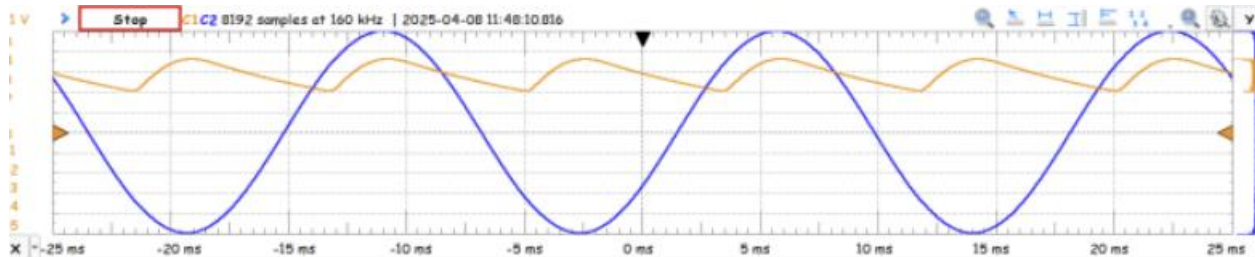


Figure 3.12: Waveforms with 10uF Capacitor

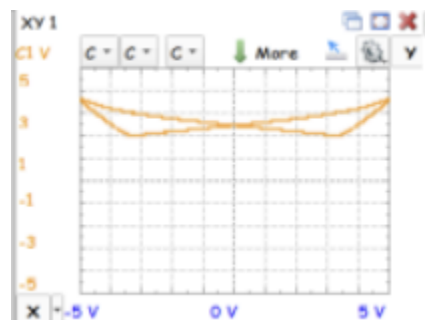


Figure 3.13: Transfer Function with a 10uF Capacitor

- 100uF Capacitor

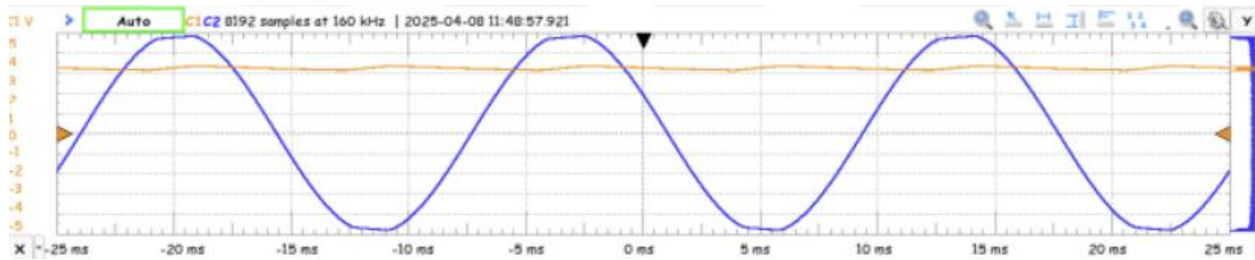


Figure 3.14: Waveform with a 100uF Capacitor



Figure 3.15: Transfer Function with a 100uF Capacitor

g. Comparing Results

The capacitor results for the full-wave rectifier closely matched the PSpice simulations. As capacitance increased, the peak-to-peak ripple voltage decreased, resulting in a smoother output.

4. Voltage Regulated Power Supply

a. Build the Circuit

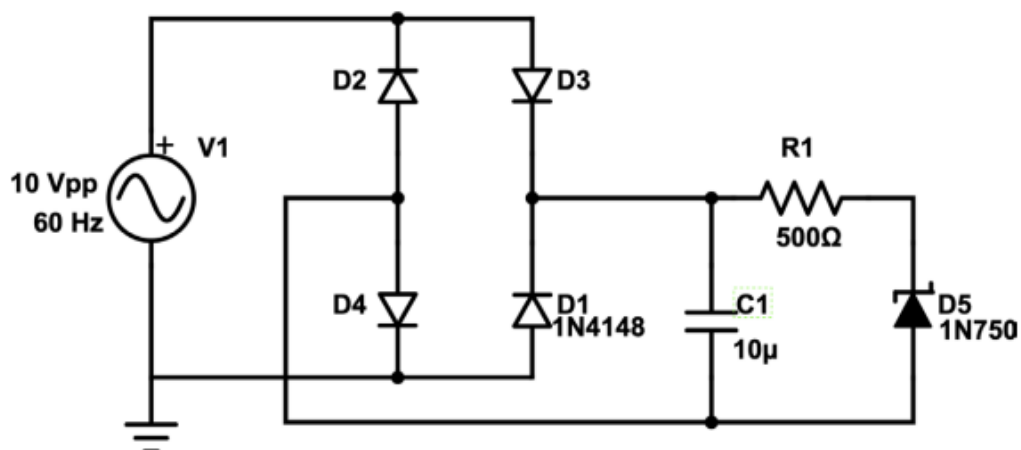


Figure 4.1: Voltage Regulated Power Supply Circuit Setup

b. Pspice Circuit Simulation and Waveform

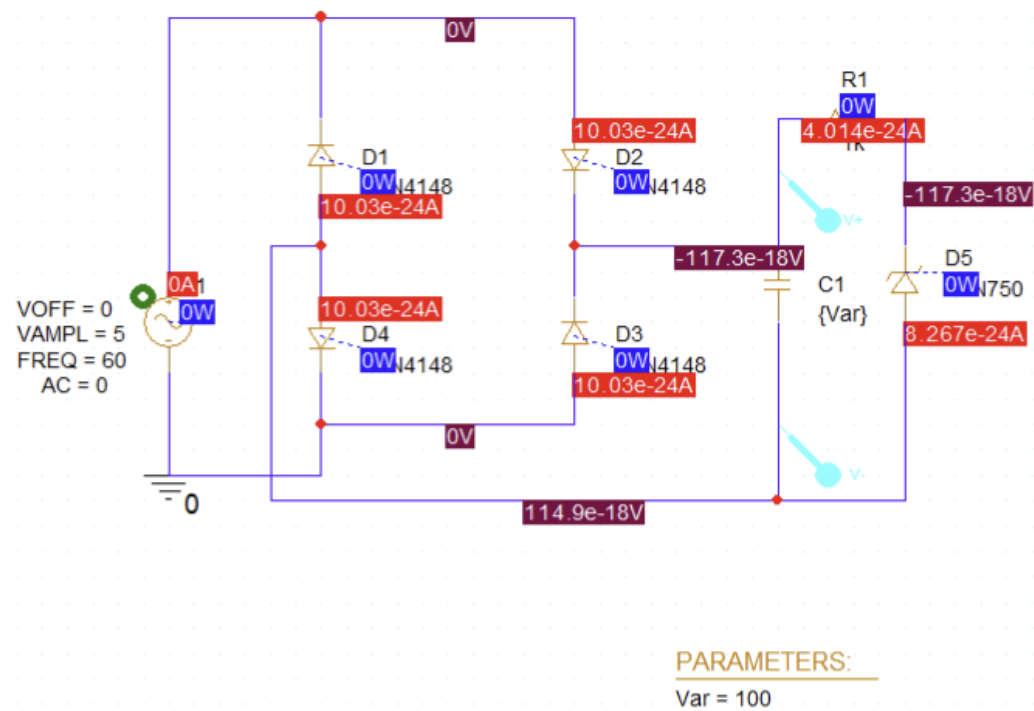


Figure 4.2: Voltage Regulated Power Supply Circuit Simulation

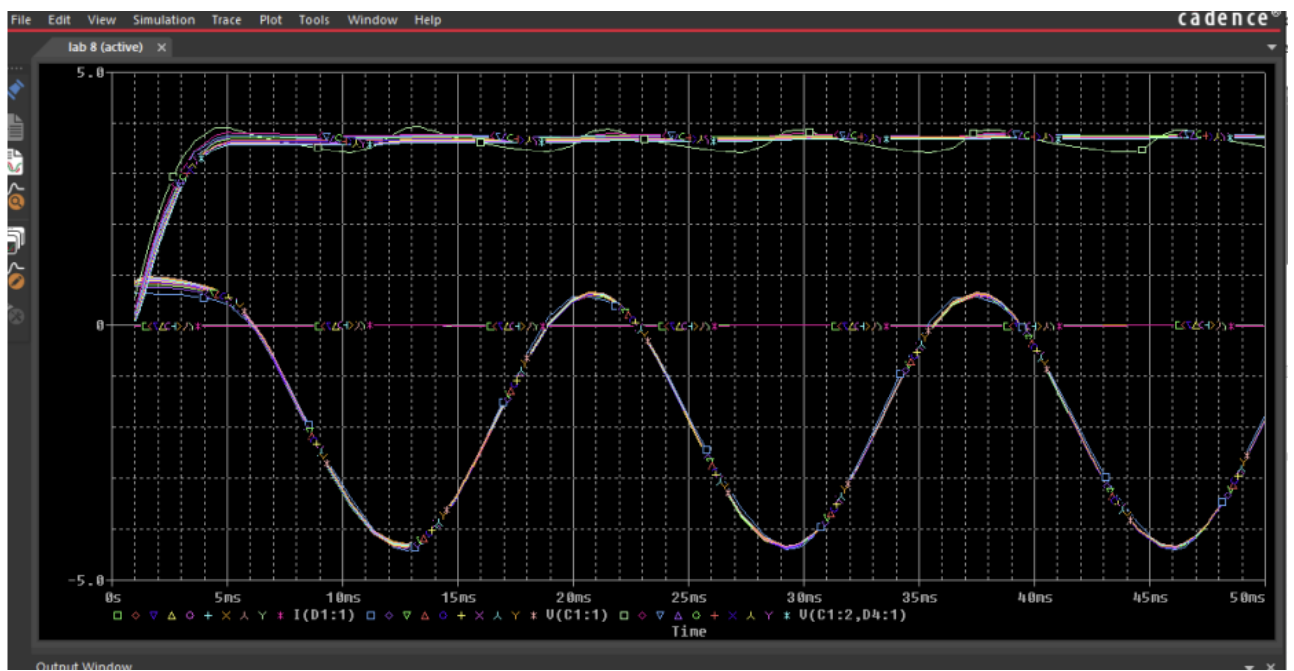


Figure 4.3: Voltage Regulated Power Supply Waveform Results

c. Physical Circuit and Simulation Results

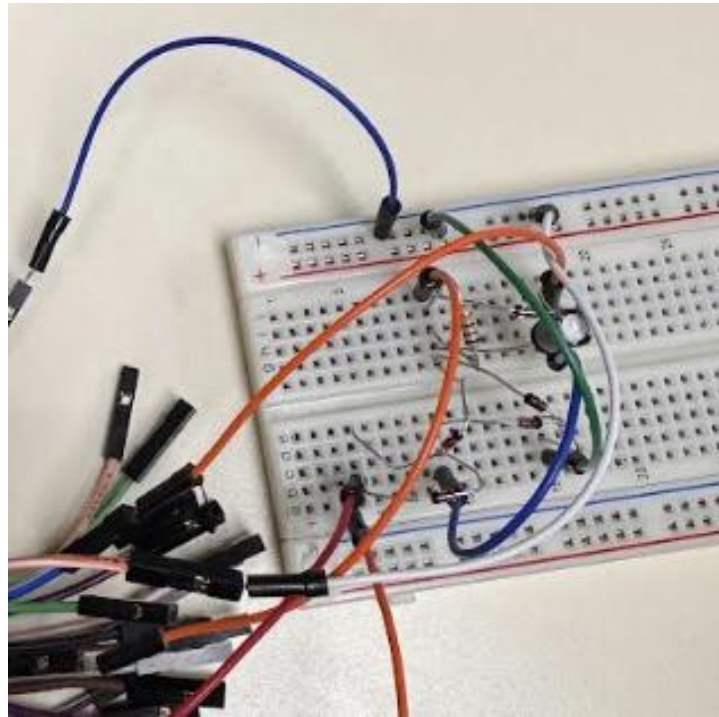


Figure 4.4: Voltage Regulated Power Supply Circuit Setup

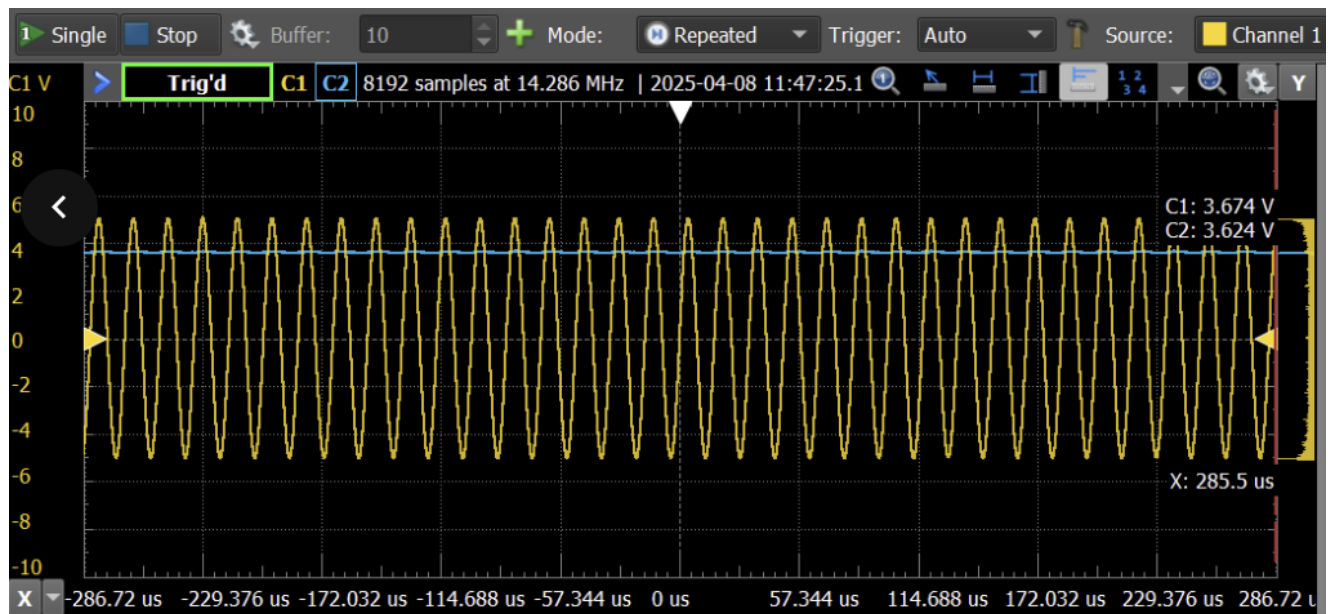


Figure 4.5: Voltage Regulated Power Supply Waveform Results

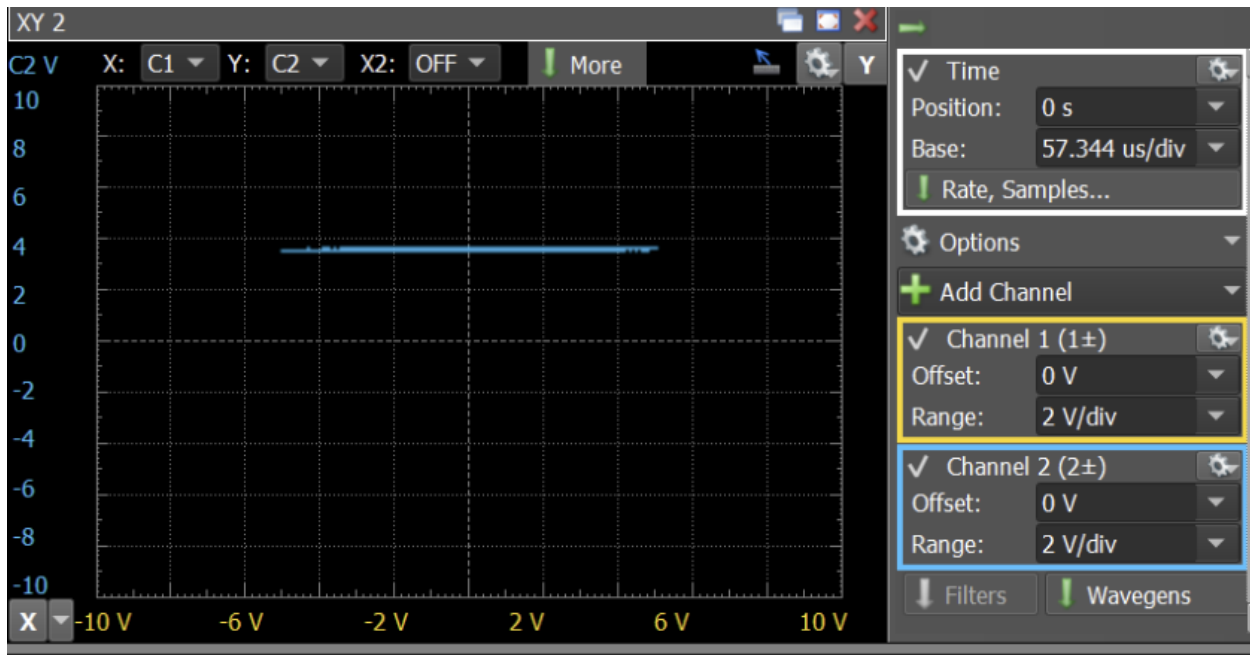


Figure 4.6: Voltage Regulated Power Supply Transfer Function

d. Comparing Results

The results from the prelab closely matched those observed in the lab. The waveforms aligned well with PSpice simulations, and the circuits performed similarly to real-world expectations. Any differences were minimal, likely due to non-ideal or aged components.

IV. Experimental Results

Each circuit in this lab was constructed and tested using both simulation and physical implementation. The results obtained from real-world measurements showed strong alignment with the expected outcomes from theoretical analysis and PSpice simulations.

For the limiting circuit, the diode successfully restricted the output voltage once the input exceeded its forward threshold of approximately 0.7V. The measured I-V characteristics and transfer function matched the predicted behavior, clearly illustrating the voltage-clipping function of the diode. Although slight differences were observed between simulation and experimental results, these were minimal and likely due to component tolerances or oscilloscope calibration.

In the half-wave rectifier circuit, the output waveform displayed a clear unidirectional pattern. The diode effectively blocked the negative half of the input AC signal, resulting in a pulsating DC output. The peak output voltage was slightly lower than the input due to the diode's forward voltage drop. The experimental waveform was consistent with PSpice simulation results, validating the circuit's performance as a basic rectifier.

The full-wave rectifier produced an output that included both halves of the AC waveform, flipped to the positive side. This confirmed the expected full-wave rectification behavior. When a capacitor was added in parallel with the load, the ripple voltage decreased noticeably. Simulations using a parametric sweep of different capacitance values were supported by the measured data, which showed that increasing the capacitance led to smoother output voltages. This demonstrated the effectiveness of filtering in reducing ripples.

Finally, the voltage-regulated power supply utilized a Zener diode to maintain a steady output voltage. Compared to the unregulated full-wave rectifier, the power supply circuit produced a much smoother DC output with significantly reduced ripple. The waveform results matched simulation predictions, and the circuit successfully demonstrated how reverse-biased Zener diodes can regulate voltage.

Overall, the experimental results confirmed the theoretical expectations and simulation outputs for each circuit. The lab provided clear evidence of how diodes operate in real-world applications and reinforced their critical role in rectification, signal conditioning, and voltage regulation.

V. Analytical Questions

1. Limiting Rectifier Circuit

- a. What does the DC sweep graph of the resistor-diode pair you measured in lab illustrate?

The graph illustrates that when a positive voltage is applied to the resistor-diode pair, the diode begins conducting once the voltage exceeds its forward threshold (0.7V), resulting in the sinusoidal waveform being clipped below this level. When the input voltage becomes negative, the diode blocks current flow due to its one-way conduction, and no current passes through.

2. Half-Wave Rectifier Circuit

- a. Why do the peak values of the input and output differ for this circuit?

The diode has a voltage drop across it, so when looking out at the peak-to-peak shape of the output and the input, it can be seen that the output is less because of the diode.

- b. Comparing the limiting circuit and the half-wave rectifier:

Both circuits alter the input waveform using the diode's one-way conduction property, but their purposes differ. The limiting circuit clips the voltage to a fixed level without producing a unidirectional signal, while the half-wave rectifier blocks one half of the AC signal, producing a pulsating DC output.

- c. Compare the 60 Hz sinusoidal wave with $V_{PP} = 10\text{ V}$ input to the outputs from the two circuits. If you combine the outputs from the two circuits, what do you get?

The limiting circuit output shows a clipped waveform, while the half-wave rectifier output shows only the positive half of the waveform. If the outputs are combined, you'd get one full sinusoidal voltage waveform.

- d. Compare the functions of the two circuits based on your observations in part 2a.

The limiting circuit protects components by capping the voltage beyond a set level, while the half-wave rectifier only sends the positive half of the signal. Both circuits are similar, one has a sinusoid above 0V and another below 0V.

3. Full-Wave Rectifier Circuit

- a. On the graph of the PSpice parameter sweep of capacitance you created in the pre-lab section; plot the data points you measured in lab. How well do the measured values compare with the plot obtained using PSpice?

The measured data points closely align with the PSpice simulation. As capacitance increased, the ripple voltage decreased in both the simulation and the experiment. Minor differences can be attributed to real-world component tolerances and measurement limitations.

4. Voltage Regulated Power Supply
 - a. How does the ripple voltage of the power supply compare to the ripple voltage of the full-wave rectifier?

The ripple voltage of the power supply is lower than that of the full-wave rectifier. This is primarily due to the addition of the Zener diode in series with the resistor, which regulates voltage by conducting in both forward and reverse bias within its breakdown region. As a result, the power supply produces a significantly smoother output with reduced ripples.

VI. Thoughtful Questions

1. What does your data indicate about your diodes' behavior according to the realistic diode model? What regions did your diode have or didn't have?

The data indicates the diodes followed the realistic model: negligible current flow in reverse bias and a forward voltage drop of approximately 0.7V in conduction. The diodes did not enter breakdown or show significant leakage current in reverse bias, suggesting they remained in their expected operating regions.

2. Determine what will happen to the full-wave rectifier circuit if:

- a. D1 is disconnected

If D1 is disconnected, one half of the AC input will not be rectified, resulting in a waveform like a half-wave rectifier during that portion of the cycle. The circuit will lose full-wave rectification.

- b. D1's polarity is reversed and everything else remains the same

Reversing D1 will cause it to conduct during the wrong half of the AC cycle, disrupting the rectification process. This may lead to incorrect or distorted output and possibly allow reverse current through the transformer windings, depending on the setup.

3. Do a PSpice simulation to verify your answer to question 2.

- a. D1 is disconnected

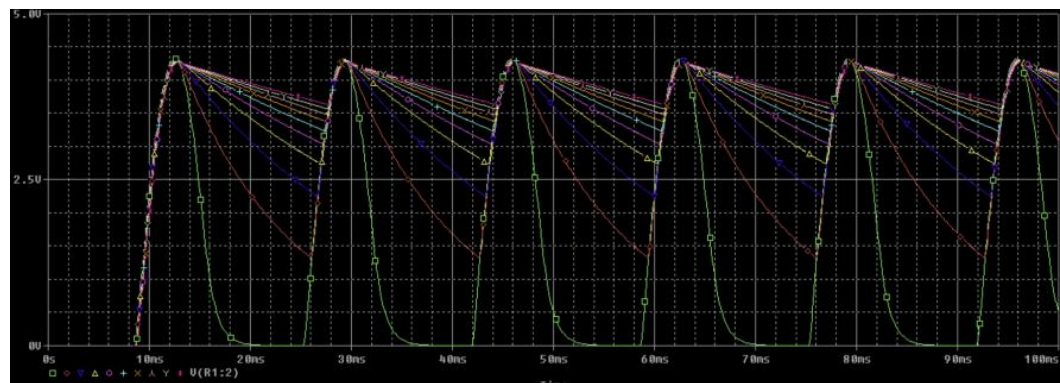


Figure 6.1: Waveform for question 2a

- b. D1's polarity is reversed

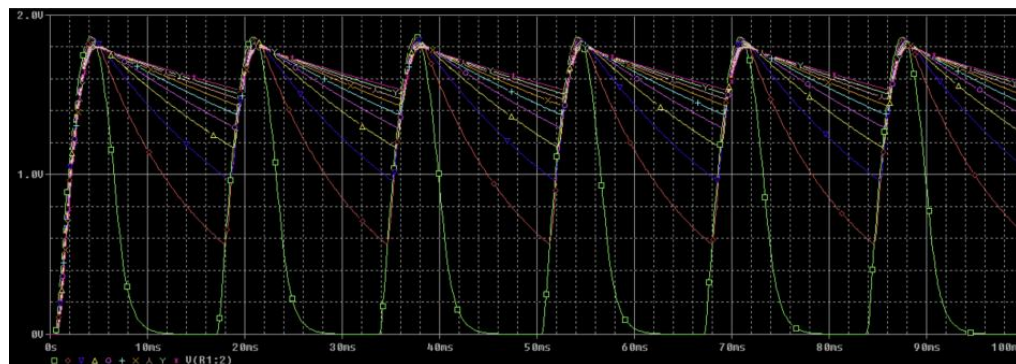


Figure 6.2: Waveform for question 2b

4. Will either or both modifications described in question 2 cause a hazardous situation? If so, which one? both? Why?

Modification 1 (disconnecting D1) is not hazardous, as it simply prevents one half of the AC cycle from being rectified, resulting in reduced functionality but no immediate risk to components. However, Modification 2 (reversing D1's polarity) can be dangerous. Reversing the diode may allow excessive reverse current, potentially damaging the diode and other nearby components. This incorrect polarity poses a greater risk and can create a hazardous situation within the circuit.

VII. Discussion and Conclusions

This lab successfully demonstrated the fundamental behaviors and practical applications of diodes in electronic circuits, emphasizing their roles in signal limiting, rectification, and voltage regulation. Simulations were first conducted using OrCAD and PSpice to model circuit behavior and visualize expected waveforms. These simulations were then verified through hands-on experiments using an AD2 connected to Waveforms, allowing for real-time observation of voltage, current, and waveform outputs from the physical circuits.

Building and analyzing the limiting circuit, half-wave rectifier, full-wave rectifier, and voltage-regulated power supply helped solidify the understanding of how diode orientation and component setup directly impact circuit functionality. Interpreting both simulated and measured data provided valuable insight into the close correlation between theoretical models and real-world performance. This lab reinforced the significance of diodes in electrical engineering and demonstrated their versatility in controlling current flow, converting AC to DC, and regulating voltage with accuracy and reliability.